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Site-specific cleavage of bacterial MucD by secreted proteases mediates antibacterial resistance in *Arabidopsis*

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Plant innate immunity restricts growth of bacterial pathogens that threaten global food security. However, the mechanisms by which plant immunity suppresses bacterial growth remain enigmatic. Here we show that *Arabidopsis thaliana* secreted aspartic protease 1 and 2 (SAP1 and SAP2) cleave the evolutionarily conserved bacterial protein MucD to redundantly inhibit the growth of the bacterial pathogen *Pseudomonas syringae*. Antibacterial activity of SAP1 requires its protease activity in planta and in vitro. Plants overexpressing *SAP1* exhibit enhanced MucD cleavage and resistance but incur no penalties in growth and reproduction, while *sap1 sap2* double mutant plants exhibit compromised MucD cleavage and resistance against *P. syringae*. *P. syringae* lacking *mucD* shows compromised growth in planta and in vitro. Notably, growth of $\Delta mucD$ complemented with the non-cleavable MucD^{F106Y} is not affected by SAP activity in planta and in vitro. Our findings identify the genetic factors and biochemical process underlying an antibacterial mechanism in plants.

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